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012

MESH REFINEMENT PROCESSES BASED ON THE GENERALIZED BISECTION OF SIMPLICES*

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Abstract. Mesh-refinement processes based on the generalized bisection of simplices are discussed and characterized in the context of the general theory of Rheinboldt [SIAM J. Numer. Anal., 17 (1980), pp. 766–778]. Then it is proved that such processes allow the construction of sequences of naturally smooth, conforming and nested nonuniform meshes. In fact it is shown that there exists a generalized mesh-refinement operator that for any conforming triangulation Δ and for any $V \subset \Delta$, produces a nested, smooth, conforming triangulation Δ^* containing successors of all elements of V and such that the minimum cell-size of Δ^* is bounded from below by half the minimum cell-size of Δ. Two conforming mesh-refinement algorithms that allow the selective refinement of computational triangulations are explicitly given.

1. Introduction. The development of adaptive and multilevel finite element solvers for elliptic boundary value problems (see e.g. [1], [2], [3], [4], [5], [8], [9], [11], [12]) depends heavily on the adequate generation of the sequences of geometrical meshes involved. Ideally, these meshes should be conforming, in the standard terminology used with finite elements, and such that the transition between large and small elements is gradual (see e.g. [5]). So far, the mesh-refinement algorithms have been based on the construction of elements similar to those of the initial mesh, and the smoothness and conformity of each mesh have been specially enforced (see e.g. [3], [5], [11]). Consequently either the computation of the stiffness matrices becomes more complex (see e.g. [11]) or the sequences of finite element spaces are not strictly nested (see e.g. [5]).

In this work we discuss another approach which is based on the generalized bisection of simplices, and which allows the construction of sequences of naturally smooth and conforming nonuniform meshes. More specifically, mesh-refinement processes based on the generalized bisection of simplices are discussed and characterized by using the general theory recently developed by Rheinboldt [10] and some known results about the generalized bisection of 2-simplices [14], [15]. Then, in this context, it is shown that there exists a generalized mesh-refinement operator ρ^* that, for any conforming triangulation Δ and for any refinement-submesh $V \subset \Delta$, produces a nested, conforming triangulation Δ^* which contains successors of all elements of V. Furthermore, the transition between small and large triangles is gradual in Δ^* and the minimum cell-size of Δ^* is bounded from below by half the minimum cell-size of Δ. In addition, two conforming mesh-refinement algorithms that allow the selective refinement of computational triangulations are explicitly discussed.

2. Generalized bisection of simplices. Consider any set $\{a_j\}_{j=1}^{n+1}$ of independent points in R^n. Then the nondegenerate n-simplex q defined by the vertices a_j, $j=1, 2, \cdots, n+1$ and denoted by $q=\langle a_1, a_2, \cdots, a_{n+1}\rangle$ will be the set [6, p. 45]:

$$(2.1)\qquad q=\left\{x=\sum_{j=1}^{n+1} \lambda_j a_j;\ 0\leqq \lambda_j \leqq 1,\ 1\leqq j\leqq n+1,\ \sum_{j=1}^{n+1} \lambda_j=1\right\}.$$

Notice that a 2-simplex is a triangle and a 3-simplex is a tetrahedron. In the same way a straight line segment can also be considered as a 1-simplex. For $n \geqq 2$, an edge or a side of the n-simplex q is any 1-simplex whose 2 vertices are also vertices of q.

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Let q be any given n-simplex. Define the diameter $\delta(q)$ by

$$\delta(q) = \max_{1 \leqq i,j \leqq n+1} \delta(\langle a_i, a_j \rangle) \quad \text{with } \delta(\langle a_i, a_j\rangle) = \|a_i - a_j\|_2. \tag{2.2}$$

If $\delta(q) = \delta(\langle a_k, a_m\rangle)$, then the side $\langle a_k, a_m\rangle$ is called a longest side of q. Additionally if $\langle a_k, a_m\rangle$ is a longest side of q and $a = (a_k + a_m)/2$, then the set $\{q_k, q_m\}$, where

$$\begin{aligned} q_k &= \langle a_1, \cdots, a_{k-1}, a, a_{k+1}, \cdots, a_m, \cdots, a_{n+1}\rangle, \\ q_m &= \langle a_1, \cdots, a_{k-1}, a_k, a_{k+1}, \cdots, a_{m-1}, a, a_{m+1}, \cdots, a_{n+1}\rangle, \end{aligned} \tag{2.3}$$

is called a generalized bisection of q [7]. If the side $\langle a_k, a_m\rangle$ is not a longest side of q, we shall say that $\{q_k, q_m\}$ is a simple bisection of q.

In what follows we shall also assume that $\delta(q) < \infty$.

3. Algebraic structure and notation. The algebraic structure associated with the generalized bisection of simplices is indeed a particular case of the structure of Rheinboldt [10].

For any n-simplex r, consider the set Q of all simplices recursively generated by the refinement operator of bisection σ such that, for any $q \in Q$, $\sigma(q) = \{q_k, q_m\} \subset Q$ is the generalized bisection of q as defined in § 2. Then the set-pair (Q, A) with $A = \{(p, q) \in Q \times Q | q \in \sigma(p)\}$ is a binary tree $\mathcal{T}$ with the root r. For any $q \in Q$, the negative and positive adjacency sets are denoted by

$$\alpha(q) = \{p \in Q/(p, q) \in A\}, \qquad \sigma(q) = \{p \in Q|(p, q) \in A\}. \tag{3.1}$$

The level $d(q)$ of any node q will be the length n of the unique path $r = p_0$, $p_1, \cdots, p_n = q$ with $(p_{i-1}, p_i) \in A$. Notice that with this definition the level of the root is 0 instead of 1 as in [10].

A refinement subset of $\mathcal{T}$ will be any finite set $R \subset Q$ such that $T(R) = (R, A(R))$ is a finite binary subtree of $\mathcal{T}$, where $A(R) = A \cap (R \times R)$. The set $\Delta = \Delta(R)$ of all terminal nodes of the binary subtree $T(R)$ is called the mesh of $\mathcal{T}$ on r generated by T. Thus, the generation of a particular mesh Δ on r is represented by the particular finite subtree of $\mathcal{T}$ which has the set of simplices of Δ as terminal nodes.

The set $\mathcal{R}$ of all refinement subsets of $\mathcal{T}$ is a distributive lattice under the set operations of union and intersection. Hence, the set $\mathcal{M}$ of all meshes $\Delta = \Delta(R)$, $R \in \mathcal{R}$ is a distributive lattice under appropriate join and meet operations (for 2-simplices, these are set operations on the set of sides of the triangles).

By introducing in $\mathcal{M}$ the partial ordering associated with the meet operation [10], namely,

$$\Delta_1 \leqq \Delta_2 \leftrightarrow \Delta_1 \wedge \Delta_2 = \Delta_1, \tag{3.2}$$

where $\Delta_1 < \Delta_2$ if $\Delta_1 \leqq \Delta_2$ and $\Delta_1 \neq \Delta_2$, we shall say that $\Delta_2 = \Delta(R_2)$ covers $\Delta_1 = \Delta(R_1)$ if $\Delta_2 > \Delta_1$ and $\Delta_2 > \Delta > \Delta_1$ does not hold for any $\Delta = \Delta(R) \subset \mathcal{M}$.

With this, we may consider maximal chains $\Delta_1 > \Delta_2 > \cdots > \Delta_m = \Delta^*$, between any two meshes $\Delta_1 = \Delta(R_1) > \Delta^* = \Delta(R^*)$ of $\mathcal{M}$ such that Δ_i covers Δ_{i+1} for $i = 1, 2, \cdots$, $m-1$. All such maximal chains have the same length $m-1$.

The meshes $\Delta(U_k)$ are called uniform meshes of $\mathcal{T}$ for

$$U_k = \{q \in Q | d(q) \leqq k\}, \qquad k = 0, 1, 2, \cdots. \tag{3.3}$$

For any mesh $\Delta = \Delta(R) \neq \Delta_r$, let $\bar{d}(\Delta)$ and $\underline{d}(\Delta)$ be defined as follows

$$\bar{d}(\Delta) = \max(d(q), q \in \Delta), \qquad \underline{d}(\Delta) = \min(d(q), q \in \Delta). \tag{3.4}$$

Obviously, $\underline{\Delta} = \Delta(U_{\underline{d}(\Delta)}) \leqq \bar{\Delta} = \Delta(U_{\bar{d}(\Delta)})$.

Finally, in line with the definition of the refinement subtrees of $\mathcal{T}$, consider the following mesh-refinement operator:

$$\rho: \text{dom}(\rho) \subset \mathcal{M} \times Q \to \mathcal{M}, \qquad \text{dom}(\rho) = \{(\Delta, q) \in \mathcal{M} \times Q | q \in \Delta\},$$
$$\rho(\Delta, q) = \Delta(R \cup \sigma(q)) \quad \forall \Delta = \Delta(R) \in \mathcal{M}, \quad q \in \Delta. \tag{3.5}$$

4. Rates of convergence. In what follows we shall restrict our analysis to 2-simplices. However, most of the results of this section can be generalized to n-simplices.

Firstly we shall define the set valued function Z which partitions the set of triangles Q into similarity classes

$$Z: Q \to W = \{\{\alpha, \beta, \gamma\}, \alpha, \beta, \gamma > 0, \alpha + \beta + \gamma = \pi\}, \tag{4.1}$$

where $Z(q) = \{\alpha, \beta, \gamma\}$ if α, β, γ are the interior angles of q.

As a measure of the degeneracy of $q \in Q$, we define the mapping $S: Q \to [0, \pi]$, such that $S(q)$ is the minimum interior angle of the triangle q.

The following two theorems characterize the mapping δ, S, Z [14], [15]:

THEOREM 1. *Let $\mathcal{T} = (Q, A)$ be the rooted binary tree associated with the generalized bisection of any nondegenerate triangle r. Then*

$$\text{(a)}\ \delta(p) \leqq (\sqrt{3}/2)^{\min(j,N)}(1/2)^{\max(j-N,0)}\delta(r) \quad \forall p \in \Delta(U_{2j}), \quad \forall j \geqq 0,$$
$$\text{(b)}\ S(q) \geqq S(r)/2 \quad \forall q \in Q, \quad q \neq r, \tag{4.2}$$

where N is a positive integer that depends on the geometry of r.

THEOREM 2. (a) *For any triangle r, the mapping Z partitions the node set Q into a finite number of similarity classes.*

(b) *Let S be the set of triangles $q = \langle a_1, a_2, a_3\rangle$ which satisfy the following relation between their sides*:

$\delta(\langle a_1, a_3\rangle) \leqq \delta(\langle a_2, a_3\rangle) \leqq \delta(\langle a_1, a_2\rangle)$, and the midpoints $a_4 = (a_1 + a_2)/2$, $a_5 = (a_2 + a_3)/2$ are such that $\delta(\langle a_1, a_3\rangle) \geqq \max\{\delta(\langle a_1, a_4\rangle), \ \delta(\langle a_3, a_4\rangle)\}$ and $\delta(\langle a_3, a_4\rangle) \geqq \delta(\langle a_3, a_5\rangle)$. Now consider $P_j(r) = \bigcup_{q \in \Delta(U_{2j}) \cap S} q$.

Then the fraction of the total area of r covered by $P_j(r)$ is a monotonically increasing sequence and $\lim_{j\to\infty} P_j(r) = r$.

Theorems 1 and 2 characterize the geometrical properties of the cells of the tree associated with the generalized bisection of any triangle r. In particular (4.2)(a) gives an upper bound over the cells of uniform meshes and (4.2)(b) assures the nondegeneracy of the triangles generated throughout the process. In addition, according to part (a) of Theorem 2, each root r yields only a finite number of similarity-distinct triangles, while part (b) of Theorem 2 assures that the shape regularity of the triangles is improved when the level increases. Part (b) of Theorem 2 is due to Rosenberg and Stenger [14], while part (a) of Theorem 1 and Theorem 2 (respectively [15, Thm. 1, Corollaries 1 and 2] are due to Stynes. In [7], Kearfott has proved an analogous property to (4.2)(a) for n-simplices. However, as far as we know, analogous results for the remaining properties have not been obtained as yet for $n > 2$.

It is worth pointing out that the remaining results of this section can be generalized to n-simplices since they only depend on (4.2)(a) and on the following property also satisfied by refinement processes involving generalized bisection of simplices, namely

$$\delta(p) \geqq \delta(\alpha(p))/2 \quad \forall p \in Q. \tag{4.3}$$

Notice that (4.3) corresponds exactly to the assumption (3.1)(ii) in [10], while on the other hand (4.2)(a) replaces the assumption (3.1)(i) in [10].

In line with the theory of Rheinboldt and as a measure of any mesh $\Delta \in \mathcal{M}$ of $\mathcal{T}$, we consider some l_μ-norm

$$\|(\delta, \Delta)\|_\mu = \begin{cases} \left(\sum_{q\in\Delta} \delta(q)^\mu \right)^{1/\mu}, & 1 \leqq \mu < \infty, \\ \max_{q\in\Delta} \delta(q), & \mu = \infty. \end{cases}$$

Clearly the trivial mesh $\Delta_r = \{r\}$ always satisfies $\|(\delta, \Delta_r)\|_\mu = \delta(r)$, $1 \leqq \mu \leqq \infty$. For the uniform meshes we have the following results:

THEOREM 3. *For any uniform mesh* $\Delta(U_{2k}) \in \mathcal{M}$ *we have*

$$\text{(4.4)} \quad \begin{aligned} &\text{(i)} \quad \|(\delta, \Delta(U_{2k}))\|_\infty \leqq (\sqrt{3}/2)^{\min(k,N)}(1/2)^{\max(k-N,0)}\delta(r) \leqq \delta(r), \forall k, \\ &\text{(ii)} \quad \|(\delta, \Delta(U_{2k}))\|_\mu \leqq (\sqrt{3})^{\min(k,N)} 2^{k(2-\mu)/\mu}\delta(r), \forall k, \mu < \infty, \\ &\text{(iii)} \quad \|(\delta, \Delta(U_{2k}))\|_\mu \leqq \delta(r), \textit{for } \mu \geqq 10, \forall k, \end{aligned}$$

where N is defined in Theorem 1.

Proof. Part (i) follows immediately from (4.2)(a) and the definition of the l_∞ norm. For the other norms, by (4.2)(a) and since any uniform mesh $\Delta(U_{2k})$ contains 2^{2k} cells, we have that

$$\text{(4.5)} \qquad \|(\delta, \Delta(U_{2k}))\|_\mu \leqq (\sqrt{3})^{\min(k,N)}\{2^{2k}(1/2)^{k\mu}\}^{1/\mu}\delta(r), \qquad \mu < \infty, \quad \forall k.$$

Hence, we immediately have (ii).

In order to prove (iii) consider the factor

$$\text{(4.6)} \qquad (\sqrt{3})^{\min(k,N)} 2^{k(2-\mu)/\mu}$$

which multiplies $\delta(r)$ in the right-hand side of (ii).

For $k < N$, the restriction $3^{k/2}2^{k(2-\mu)/\mu} \leqq 1$ implies $\mu \geqq 2/(1-C/2)$, where $C = \log 3/\log 2$.

Since $2/(1-C/2) < 10$, the inequality (iii) immediately follows for $\mu \geqq 10$, $k < N$.

By analogy, if we require that (4.6) is less than 1 for $k \geqq N$, we obtain the condition $\mu \geqq 2k/(k-NC/2)$, and since for $k \geqq N$, the function $2k/(k-NC/2)$ is monotonically decreasing, it follows that

$$\frac{2k}{k-NC/2} \leqq \frac{2N}{N(1-C/2)} = \frac{2}{(1-C/2)} < 10 \quad \forall k,$$

which implies (iii) for $\mu \geqq 10$, $k \geqq N$.

For the other meshes we have the following results:

THEOREM 4. (a) *For any nonuniform mesh* $\Delta \in \mathcal{M}$,

$$\text{(4.7)} \quad \begin{aligned} &\text{(i)} \quad \|(\delta, \Delta)\|_\infty \leqq (\sqrt{3}/2)^{\min(\tilde{d},N)}(1/2)^{\max(\tilde{d}-N,0)}\delta(r) \leqq \delta(r), \\ &\text{(ii)} \quad \|(\delta, \Delta)\|_\mu \leqq \delta(r) 3^{N/2} \sum_{k=\underline{d}}^{\bar{d}} |\Delta \cap \Delta(U_k)|^{1/\mu} 2^{-(k-1)/2}, \qquad \mu < \infty. \end{aligned}$$

(b) *For any integer* $M > 0$, *there exist* $D > 0$ *and* $\bar{\mu} > 0$ *such that if* $\Delta \in \mathcal{M}$ *satisfies* $\underline{d}(\Delta) \geqq D + 1$ *and* $(\bar{d} - \underline{d}) \leqq M - 1$ *then*

$$\text{(4.8)} \qquad \|(\delta, \Delta)\|_\mu \leqq \delta(r) \quad \textit{for } \bar{\mu} \leqq \mu < \infty,$$

where $D = 1 + (N \log 3 + 2 \log M)/\log 2$ *and* $\bar{\mu} \geqq 2(D+1)$, $\underline{d} = \underline{d}(\Delta)$ *and* $\bar{d} = \bar{d}(\Delta)$ *are defined as in* (3.4), *N is defined in Theorem* 1 *and* $\tilde{d}$ *is the integer part of* $\underline{d}/2$.

Proof. From the definition of the l_∞ norm and property (4.2)(a), the inequality (4.7)(i) immediately follows for $\mu = \infty$.

In order to prove (4.7)(ii) we use the definition of the l_μ norm and the property (4.2)(a) to get

$$\|(\delta, \Delta)\|_\mu \leqq \delta(r)\left(\sum_{k=\underline{d}}^{\bar{d}} \sum_{q\in\Delta\cap\Delta(U_k)} 3^{\mu\min(\tilde{k},N)/2}2^{-\tilde{k}\mu}\right)^{1/\mu}, \tag{4.9}$$

where we have partitioned the mesh in accordance with the level of its cells and $\tilde{k}$ is the integer part of $k/2$, for $\underline{d} \leqq k \leqq \bar{d}$. From (4.9) it follows that

$$\begin{aligned}\|(\delta, \Delta)\|_\mu &\leqq \delta(r) \sum_{k=\underline{d}}^{\bar{d}} 3^{\min(\tilde{k},N)/2}\left(\sum_{q\in\Delta\cap\Delta(U_k)} 2^{-\tilde{k}\mu}\right)^{1/\mu} \\ &\leqq \delta(r)3^{(\max_k\{\min(\tilde{k},N)/2\})} \sum_{k=\underline{d}}^{\bar{d}} |\Delta\cap\Delta(U_k)|^{1/\mu}2^{-\tilde{k}} \\ &\leqq \delta(r)3^{N/2} \sum_{k=\underline{d}}^{\bar{d}} |\Delta\cap\Delta(U_k)|^{1/\mu}2^{-(k-1)/2},\end{aligned} \tag{4.10}$$

which proves (4.7)(ii) for $\mu<\infty$.

Since the cardinality $|\Delta\cap\Delta(U_K)|$ can be bounded by $|\Delta\cap\Delta(U_k)|\leq|\Delta(U_K)|=2^k$, (4.8) holds if

$$3^{N/2}2^{k/\mu-(k-1)/2} \leqq 1/(\bar{d}-\underline{d}+1) \quad \text{for } \underline{d}\leqq k\leqq k\leqq\bar{d}, \tag{4.11}$$

which implies the following condition on μ:

$$\mu \geqq 2k/(k-\hat{D}) \quad \text{for } \underline{d}\leqq k\leqq \bar{d}, \tag{4.12}$$

and

$$\hat{D} = 1+(N\log 3+2\log(\bar{d}-\underline{d}+1))/\log 2.$$

Notice that the right-hand side of (4.12) is a monotonically decreasing function of k for $k>\hat{D}$. Thus if we select any fixed integer $M>0$, (4.8) holds for each mesh Δ such that $(\bar{d}-\underline{d})\leqq M-1$, $\underline{d}(\Delta)\geqq D+1$, and for $\mu\geqq\bar{\mu}\geqq 2(D+1)$, where $D=1+(N\log 3+2\log M)/\log 2$.

The conditions $\underline{d}\geqq D+1$ and $(\bar{d}-\underline{d})\leqq M-1$ are not so restrictive in practice, since by selecting a reasonable value for M we cover a large set of meshes.

Notice that Theorems 3 and 4 replace the first part of [10, Thm. 3]. A result analogous to the second part of [10, Thm. 3] can be obtained for the generalized bisection of simplices in the case $\mu=1$.

As in [10], for the control of the refinement process we need a given error indicator function $\varepsilon: Q\to\{t\in R^1 | t\geqq 0\}$ which is monotone with respect to refinement, namely such that $\varepsilon(p)\leqq\varepsilon(\alpha(p))$, $\forall p\in Q$, $p\neq r$, and the error of any mesh $\Delta\in\mathcal{M}$ is measured by some l_ν norm for $1\leqq\nu\leqq\infty$.

In order to construct meshes with optimal accuracy within a prescribed cost range (see e.g. [3], [12]), a mesh $\Delta\in\mathcal{M}$, $\Delta\neq\Delta_r$, will be called ε-optimal if $\max_{p\in\Delta}\varepsilon(p)\leqq\min_{p\in\Delta}\varepsilon(\alpha(p))$.

Thus for ε monotone, a mesh Δ will be ε-optimal if $\varepsilon(p)=\text{constant}$ for all $p\in\Delta$. [10, Thm. 5] which characterizes refinement strategies with ε-optimality is also valid for mesh-refinements by generalized bisection of simplices. Therefore, the two mesh-refinement algorithms described in [10] can also be used for constructing sequences of ε-optimal meshes in this particular case.

Finally, by assuming that $\varepsilon(p)\leqq\beta\delta(p)^k$, $\forall p\in Q, \beta, k>0$, we obtain results analogous to those of [10, Thms. 7, 8] with respect to the rate of change of the errors of the meshes as a function of their cardinality $|\Delta|$. Global optimality properties of the type discussed in § 4 in [10] can be also proved for this particular kind of mesh-refinement process.

5. Sequences of conforming meshes. In what follows we show that, by using generalized bisection of 2-simplices, sequences of nested, smooth and conforming triangulations can be generated, even if local refinement is used. Some of these ideas have been discussed from a more practical point of view in [13].

Firstly, for any $\Delta = \Delta(R) \in \mathcal{M}$, we introduce the following definitions:

DEFINITIONS. A triangle $p \in \Delta$ is conforming on Δ if, for each $q \in \Delta$, $p \cap q$ is either empty, or a vertex of p, or a side of p.

A mesh Δ is conforming if each $p \in \Delta$ is conforming on Δ. Additionally we shall say that $p \in \Delta$ is a $\frac{1}{2}$=nonconforming triangle on Δ if there exist a pair of cells $q_1, q_2 \in \Delta$ such that

$$q_1 \cap p \neq \emptyset, \quad q_2 \cap p \neq \emptyset, \quad \alpha(q_1) = \alpha(q_2) = \{q\},$$

and the triangle p is conforming on $\Delta(R \backslash \sigma(q))$. Moreover we shall say that the side $m = p \cap q$ is a $\frac{1}{2}$-nonconforming side of p, and that the midpoint of m is a $\frac{1}{2}$-nonconforming point of p. Finally Δ will be a $\frac{1}{2}$-nonconforming mesh if each $p \in \Delta$ is either conforming or $\frac{1}{2}$-nonconforming on Δ.

In order to illustrate these concepts see Fig. 1 where p represents a $\frac{1}{2}$-nonconforming triangle on $\Delta = \{p, q_1, q_2\}$, and $\alpha(q_1) = \alpha(q_2) = \{q\}$ with $q = q_1 \cup q_2$. Notice that according to the definition the triangles q_1 and q_2 are conforming on Δ.

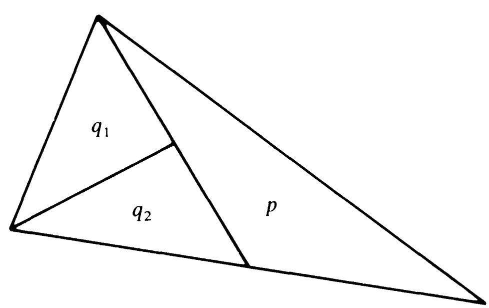

FIG 1

With the above definitions the following theorem holds.

THEOREM 5. *Let $\mathcal{T} = (Q, A)$ be the binary tree associated with the refinement by generalized bisection of any nondegenerate triangle r. Then, for any conforming triangulation $\Delta = \Delta(R) \in \mathcal{M}$ and for any $V_1 \subset \Delta$, there exists a new conforming triangulation $\Delta^* = \Delta^*(R^*) \in \mathcal{M}$ such that*

$$\text{(5.1)} \quad \begin{aligned} &\text{(a)} \quad \Delta < \Delta^*, \\ &\text{(b)} \quad \sigma(q) \subset R^*, \forall q \in V_1, \\ &\text{(c)} \quad \min_{p \in \Delta^*} \delta(p) \geqq \min\{\tfrac{1}{2}\min_{q \in V_1} \delta(q), \min_{q \in \Delta} \delta(q)\}. \end{aligned}$$

Proof. The existence of a conforming triangulation Δ is guaranteed since the uniform mesh $\Delta(U_1)$ is conforming according to the definition.

The proof of the existence of Δ^* is constructive. Set $R_1 = R$ such that $\Delta = \Delta(R)$. We shall show that it is possible to construct a finite chain $\{\Delta_j\}_{j=1}^n$ such that

$$\Delta = \Delta_1 < \Delta_2 < \cdots < \Delta_n = \Delta^*, \tag{5.2}$$

where $\{\Delta_j\}_{j=2}^{n-1}$ are $\frac{1}{2}$-nonconforming triangulations.

Define $\Delta_2 = \Delta(R_2)$ with $R_2 = R_1 \cup \bigcup_{q \in V_1} \sigma(q)$.

By [10, Lemma 2], there exists a finite maximal chain $\{\tilde{\Delta}_j\}_{j=1}^{m+1}$ such that

$$\Delta_1 = \tilde{\Delta}_1 < \tilde{\Delta}_2 < \tilde{\Delta}_3 < \cdots < \tilde{\Delta}_{m+1} = \Delta_2, \tag{5.3}$$

where $\tilde{\Delta}_{i+1}$ covers $\tilde{\Delta}_i$ for $i = 1, 2, \cdots, m$ and $m = |V_1|$.

Since Δ_1 is conforming, Δ_2 will be either a conforming or a $\frac{1}{2}$-nonconforming triangulation. If Δ_2 is conforming, then obviously $\Delta^* = \Delta_2$. If Δ_2 is a $\frac{1}{2}$-nonconforming triangulation, then consider the set $V_2 \subset \Delta_2$ of all the $\frac{1}{2}$-nonconforming triangles of Δ_2 and define

$$\Delta_3 = \Delta(R_3), \quad \text{where } R_3 = R_2 \cup \bigcup_{q \in V_2} \sigma(q).$$

By applying the above arguments to Δ_2, it can be shown that there exists a finite maximal chain of length $|V_2|$ such that the last triangulation is equal to Δ_3. The triangulation Δ_3 is either conforming or $\frac{1}{2}$-nonconforming since by definition, after a bisection of any $\frac{1}{2}$-nonconforming triangle $p \in \Delta_2$, either the triangles of $\sigma(p)$ are conforming on Δ_3 or another $\frac{1}{2}$-nonconforming triangle has been generated on Δ_3. In the latter case we say that the $\frac{1}{2}$-nonconforming refinement of p has been propagated. To illustrate this, see Fig. 2 where by refining the $\frac{1}{2}$-nonconforming triangle p, the $\frac{1}{2}$-nonconforming triangle t was generated.

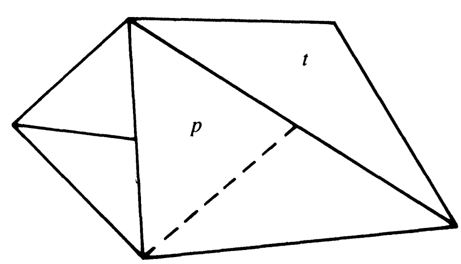

FIG. 2

Using the same reasoning, we continue iteratively until the conforming triangulation Δ^* is obtained.

In order to see that the chain (5.2) is finite, let $\bar{\mathcal{M}} \subset \mathcal{M}$ be the set of all the $\frac{1}{2}$-nonconforming triangulations and define the mapping $w: \bar{\mathcal{M}} \to \{t \in R^1 | t > 0\}$ in such a way that $w(\Delta)$ is the length of the smallest $\frac{1}{2}$-nonconforming side of Δ.

Let $\Delta_2, \Delta_3, \cdots$ be the sequence of the $\frac{1}{2}$-nonconforming meshes of (5.2). Since the refinement of any $\frac{1}{2}$-nonconforming triangle p will be propagated only if the length of its $\frac{1}{2}$-nonconforming triangle p will be propagated only if the length of its $\frac{1}{2}$-nonconforming side is less than $\delta(p)$, we have that

$$w(\Delta_j) \geqq w(\Delta_{j-1}) \quad \text{for } j > 2, \qquad w(\Delta_2) \geqq \min_{q \in V_1} \delta(q), \tag{5.4}$$

which assures the finiteness of the chain (5.2).

Finally from (5.4) follows that the length of the smallest side of Δ bisected throughout all the process is equal to $\min_{q \in V_1} \delta(q)$, which immediately implies (5.1)(c).

Theorem 5 indeed defines an algorithm for refining globally or locally any conforming triangulation Δ, in such a way that the final triangulation will be also conforming. The algorithm also ensures that all angles in subsequent refined triangulations are greater than or equal to half the smallest angle of the root r; the shape regularity of all triangles is maintained and the transition between small and large triangles is assured in a natural way (see also [13]).

Let Δ be a conforming triangulation. Then, for any refinement submesh $V \subset \Delta$, the mesh refinement algorithm can be summarized as follows.

(5.5)

1. Input $\{\Delta, R, V\}$
2. $R_1 := R$; $\Delta_1 := \Delta$; $V_1 := V$; $i := 1$;
3. While $|V_i| > 0$ do
   3.1. $R_{i+1} := R_i \cup \bigcup_{q \in V_i} \sigma(q)$
   3.2. $\Delta_{i+1} = \Delta(R_{i+1})$
   3.3. Select the set V_{i+1} of $\frac{1}{2}$-nonconforming triangles of Δ_{i+1}
   3.4. $i := i+1$

It is worth pointing out here that property (5.1)(c) is in a certain sense an extension of (4.3) for adjacent conforming meshes.

Moreover, note that in this section we have indeed defined a generalized mesh-refinement operator ρ^* such that

$$\rho^* : \mathrm{dom}\,(\rho^*) \subset \mathcal{M}_c \times \mathcal{P}(Q) \to \mathcal{M}_c,$$

$$\mathrm{dom}\,(\rho^*) = \{(\Delta, V) \in \mathcal{M}_c \times \mathcal{P}(Q) \mid V \subset \Delta\}$$

where $\mathcal{M}_c \subset \mathcal{M}$ is the set of conforming triangulations, $\mathcal{P}(Q)$ is the power set of Q, and the way of generating $\rho^*(\Delta, V)$ is described by the procedure (5.5).

With these ideas in mind, some further mesh-refinement algorithms can be constructed. Thus, for instance, the procedure (5.5) can be modified in order to obtain more practical and useful algorithms which maintain all the properties of the latter [13].

In this connection we shall describe briefly a procedure which combines generalized and simple bisections of the triangles, in such a way that, each refined triangle of Δ embeds only 2, 3 or 4 triangles of the final conforming triangulation Δ^*. To be explicit, each triangle $p \in \Delta$, refined throughout this procedure, is firstly bisected by a longest side generating two triangles p_1 and p_2 (see Fig. 3); if in any subsequent stage of the process which generates Δ^*, either the points P or Q becomes $\frac{1}{2}$-nonconforming, then either the triangle p_1 or the triangle p_2 is respectively refined by a simple bisection of its $\frac{1}{2}$-nonconforming side. Notice that the simple bisection of at least one of these triangles coincides with a generalized bisection.

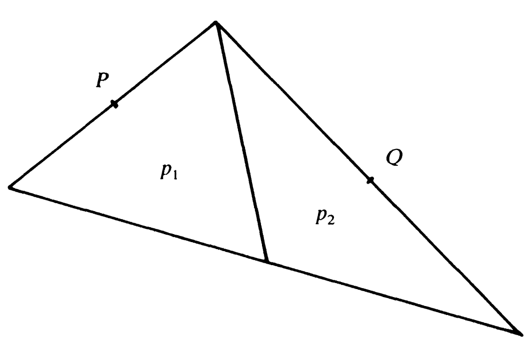

FIG. 3

The procedure can be described in the following schematic form:

(5.6)

1. Input $\{\Delta, R, V\}$
2. $R_1 := R$; $\Delta_1 := \Delta$, $V_1 := V$; $W_1 := \bigcup_{p \in V} p$; $i := 1$;

3. While $|V_i|>0$ do
   3.1. $\tilde{R}_i := R_i \cup \bigcup_{q\in V_i} \sigma(q)$
   3.2. $\tilde{\Delta}_i = \Delta(\tilde{R}_i)$
   3.3. Select the set $\tilde{W}_i$ of $\frac{1}{2}$-nonconforming triangles p of $\tilde{\Delta}_i$ and such that $p \subset W_i$
   3.4. $R_{i+1} := \tilde{R}_i \cup \bigcup_{q\in \tilde{W}_i} \tilde{\sigma}(q)$
   where $\tilde{\sigma}(q)$ is the simple bisection of q by its $\frac{1}{2}$-nonconforming side
   3.5. $\Delta_{i+1} = \Delta(R_{i+1})$
   3.6. Select the set V_{i+1} of $\frac{1}{2}$-nonconforming triangles of Δ_{i+1}
   3.7. $W_{i+1} = W_i \cup \bigcup_{p\in V_{i+1}} p$
   3.8. $i := i+1$

It is worth pointing out here that, at each stage of the procedure (5.6), $V_{i+1} \subset \Delta$ and V_{i+1} is only formed of $\frac{1}{2}$-nonconforming triangles obtained by propagation.

In order to show that the meshes generated by means of the procedure (5.6) satisfy properties analogous to those of the meshes generated by using the procedure (5.5), consider the Fig. 4 where the triangles $\tilde{p}_1 = p_1 \cup p_2$ and $\tilde{p}_2 = p_3 \cup p_4$ were generated by generalized bisection of triangle $p = p_1 \cup p_2 \cup p_3 \cup p_4$; the triangles p_3 and p_4 were generated by generalized bisection of triangle $\tilde{p}_2$; and the triangles p_1 and p_2 were obtained by simple bisection of $\tilde{p}_1$.

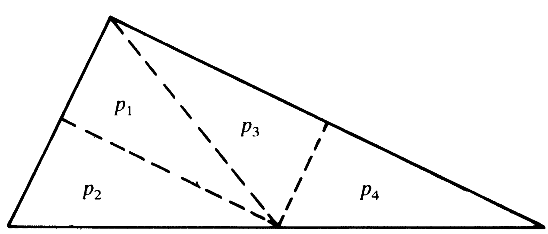

FIG. 4

It should be evident from Fig. 4 that, even if the simple bisection of $\tilde{p}_1$ does not coincide with a generalized bisection of this triangle, the triangles p_1 and p_3, and the triangles p_2 and p_4 are congruently equivalent. Thus, since the predecessor of each pair of triangles obtained by simple bisection in the procedure (5.6) is always obtained by generalized bisection of a preceding triangle, we can associate to each triangle of the final conforming mesh Δ^*, a unique level k corresponding to the level of its congruently equivalent triangle $q \in \Delta(U_k)$. For these reasons, properties analogous to those obtained for the nonuniform meshes of § 4, can be proved for this special kind of mesh. In addition, Theorem 5 of this section can also be proved by using the procedure (5.6) to generate the final conforming triangulation Δ^*.

The procedure (5.6) indeed constitutes a simple and effective tool for the selective refinement of conforming computational triangulations, a problem which had not been satisfactorily solved so far. An extensive practical discussion of the algorithms presented here, as well as some examples, can be found in reference [13]. It is worth noting, however, that the most promising application of these algorithms resides in the implementation of fully adaptive multigrid finite element solvers. In this context, the procedure (5.6), a suitable data structure, a multigrid algorithm, and a selfadaptive strategy based on the approach of Babuska and Rheinboldt (see e.g. [1], [2], [3]) have been combined in an experimental partial differential equations solver, whose preliminary evaluation has shown to be promising. More details concerning the data structure, the design and the results of this software will be published elsewhere.

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